Inverse Function Theorem: Detailed Proof from the Special Case to the General Case

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Main Theorem

Theorem 1 (Inverse Function Theorem)**.** *Let*

$$f : U \subset \mathbb{R}^n \to \mathbb{R}^n$$

be a C^1 function, where U is open. Suppose

$$p \in U$$

and the Jacobian matrix

$$Df(p)$$

is invertible. Then there exist neighborhoods

$$V \subset U \quad \text{of } p$$

and

$$W \subset \mathbb{R}^n \quad \text{of } f(p)$$

such that

$$f : V \to W$$

is one-to-one and onto. Therefore a local inverse

$$f^{-1} : W \to V$$

exists. Moreover, f^{-1} is C^1, and for every $y \in W$,

$$D(f^{-1})(y) = \big(Df(f^{-1}(y))\big)^{-1}.$$

Part I: The Special Case

First we prove the theorem under the special assumptions

$$p = 0, \qquad f(0) = 0, \qquad Df(0) = I.$$

So we assume

$$f : U \subset \mathbb{R}^n \to \mathbb{R}^n$$

is C^1, $0 \in U$, and

$$f(0) = 0, \qquad Df(0) = I.$$

Because U is open and contains 0, there exists some radius $R > 0$ such that

$$\overline{B(0, R)} \subset U.$$

Step 1: Define the error map

Define

$$g(x) = x - f(x).$$

The reason for this definition is that, since

$$Df(0) = I,$$

the function f behaves approximately like the identity map near 0. Therefore

$$g(x) = x - f(x)$$

measures the error between f and the identity map.

Now compute $g(0)$:

$$g(0) = 0 - f(0) = 0 - 0 = 0.$$

Also compute the derivative:

$$Dg(x) = D(x) - Df(x) = I - Df(x).$$

Therefore at 0,

$$Dg(0) = I - Df(0) = I - I = 0.$$

Because f is C^1, the derivative Df is continuous. Since

$$Dg(x) = I - Df(x),$$

and I is constant, Dg is also continuous.

Since

$$Dg(0) = 0,$$

and Dg is continuous, we may choose $0 < \delta < R$ so small that for every

$$x \in \overline{B(0, \delta)},$$

we have

$$\|Dg(x)\| \leq \frac{1}{2}.$$

Here $\|Dg(x)\|$ means the operator norm of the matrix $Dg(x)$.

Step 2: Prove that g shrinks distances

We now prove that for any

$$x_1, x_2 \in \overline{B(0,\delta)},$$

we have

$$\|g(x_1) - g(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\|.$$

Take two points

$$x_1, x_2 \in \overline{B(0,\delta)}.$$

Connect them by the straight line segment

$$\gamma(t) = x_2 + t(x_1 - x_2), \qquad 0 \leq t \leq 1.$$

Check the endpoints:

$$\gamma(0) = x_2,$$

and

$$\gamma(1) = x_2 + (x_1 - x_2) = x_1.$$

Because the closed ball $\overline{B(0,\delta)}$ is convex, the whole line segment from x_2 to x_1 stays inside $\overline{B(0,\delta)}$. Therefore

$$\gamma(t) \in \overline{B(0,\delta)}$$

for all $0 \leq t \leq 1$.

Now define a one-variable vector-valued function

$$h(t) = g(\gamma(t)).$$

Then

$$h(0) = g(\gamma(0)) = g(x_2),$$

and

$$h(1) = g(\gamma(1)) = g(x_1).$$

Therefore

$$g(x_1) - g(x_2) = h(1) - h(0).$$

By the Fundamental Theorem of Calculus,

$$h(1) - h(0) = \int_0^1 h'(t)\,dt.$$

Thus

$$g(x_1) - g(x_2) = \int_0^1 h'(t)\,dt.$$

Now compute $h'(t)$. Since

$$h(t) = g(\gamma(t)),$$

we use the chain rule:

$$h'(t) = Dg(\gamma(t))\gamma'(t).$$

But

$$\gamma(t) = x_2 + t(x_1 - x_2),$$

so

$$\gamma'(t) = x_1 - x_2.$$

Therefore

$$h'(t) = Dg(\gamma(t))(x_1 - x_2).$$

Hence

$$g(x_1) - g(x_2) = \int_0^1 Dg(\gamma(t))(x_1 - x_2)\, dt.$$

Take norms:

$$\|g(x_1) - g(x_2)\| \leq \int_0^1 \|Dg(\gamma(t))(x_1 - x_2)\|\, dt.$$

Using the operator norm inequality

$$\|Av\| \leq \|A\|\|v\|,$$

we get

$$\|g(x_1) - g(x_2)\| \leq \int_0^1 \|Dg(\gamma(t))\|\, \|x_1 - x_2\|\, dt.$$

Since $\gamma(t)$ stays inside $\overline{B(0,\delta)}$, and on that ball

$$\|Dg(\gamma(t))\| \leq \frac{1}{2},$$

we get

$$\|g(x_1) - g(x_2)\| \leq \int_0^1 \frac{1}{2}\|x_1 - x_2\|\, dt.$$

The quantity $\|x_1 - x_2\|$ is constant with respect to t, so

$$\|g(x_1) - g(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\| \int_0^1 dt.$$

Since

$$\int_0^1 dt = 1,$$

we conclude

$$\|g(x_1) - g(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\|.$$

So g is a contraction on $\overline{B(0,\delta)}$.

Step 3: A useful special estimate

Put

$$x_2 = 0, \qquad x_1 = x.$$

Since $g(0) = 0$, the previous estimate gives

$$\|g(x) - g(0)\| \leq \frac{1}{2}\|x - 0\|.$$

Therefore

$$\|g(x)\| \leq \frac{1}{2}\|x\|.$$

In particular, if

$$\|x\| \leq \delta,$$

then

$$\|g(x)\| \leq \frac{1}{2}\delta.$$

Thus

$$g(\overline{B(0,\delta)}) \subset \overline{B(0,\delta/2)}.$$

Step 4: Convert solving $f(x) = y$ into a fixed point problem

We want to solve

$$f(x) = y$$

for x, where y is near 0.

Recall that

$$g(x) = x - f(x).$$

Rearrange this formula:

$$g(x) = x - f(x).$$

Add $f(x)$ to both sides:

$$g(x) + f(x) = x.$$

Subtract $g(x)$ from both sides:

$$f(x) = x - g(x).$$

Therefore the equation

$$f(x) = y$$

becomes

$$x - g(x) = y.$$

Add $g(x)$ to both sides:

$$x = y + g(x).$$

Now define the map

$$g_y(x) = y + g(x).$$

Then solving

$$f(x) = y$$

is equivalent to finding a fixed point of g_y:

$$x = g_y(x).$$

The equivalence is important:

- If $f(x) = y$, then $g(x) = x - f(x) = x - y$, so $x = y + g(x) = g_y(x)$.
- If $x = g_y(x)$, then $x = y + g(x)$, so $x - g(x) = y$. But $x - g(x) = f(x)$. Therefore $f(x) = y$.

Thus solving $f(x) = y$ is exactly the same as finding a fixed point of g_y.

Step 5: Show that g_y maps the ball into itself

Now assume

$$\|y\| \leq \frac{\delta}{2}.$$

This means y is in a small output neighborhood of 0. This is allowed because the inverse function theorem is local: we only need to solve $f(x) = y$ for y sufficiently close to 0.

Take any

$$x \in \overline{B(0, \delta)}.$$

Then

$$\|x\| \leq \delta.$$

We estimate

$$\|g_y(x)\|.$$

Using the definition

$$g_y(x) = y + g(x),$$

we have

$$\|g_y(x)\| = \|y + g(x)\|.$$

By the triangle inequality,

$$\|y + g(x)\| \leq \|y\| + \|g(x)\|.$$

We know

$$\|y\| \leq \frac{\delta}{2}$$

and from the previous estimate

$$\|g(x)\| \leq \frac{1}{2}\|x\| \leq \frac{\delta}{2}.$$

Therefore

$$\|g_y(x)\| \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta.$$

Thus

$$g_y(x) \in \overline{B(0, \delta)}.$$

Since this holds for every $x \in \overline{B(0, \delta)}$, we have

$$g_y(\overline{B(0, \delta)}) \subset \overline{B(0, \delta)}.$$

Step 6: Show that g_y is a contraction

Take

$$x_1, x_2 \in \overline{B(0, \delta)}.$$

Then

$$g_y(x_1) = y + g(x_1),$$

and

$$g_y(x_2) = y + g(x_2).$$

Subtract:

$$g_y(x_1) - g_y(x_2) = \big(y + g(x_1)\big) - \big(y + g(x_2)\big).$$

The y terms cancel:

$$g_y(x_1) - g_y(x_2) = g(x_1) - g(x_2).$$

Taking norms,

$$\|g_y(x_1) - g_y(x_2)\| = \|g(x_1) - g(x_2)\|.$$

But from Step 2,

$$\|g(x_1) - g(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\|.$$

Therefore

$$\|g_y(x_1) - g_y(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\|.$$

So g_y is a contraction on $\overline{B(0,\delta)}$.

Step 7: Apply the Banach Fixed Point Theorem

The closed ball

$$\overline{B(0,\delta)}$$

is complete because it is a closed subset of the complete metric space $\mathbb{R}^n$.

We have shown:

1. g_y maps $\overline{B(0,\delta)}$ into itself.
2. g_y is a contraction on $\overline{B(0,\delta)}$.

Therefore, by the Banach Fixed Point Theorem, there exists a unique point

$$x \in \overline{B(0,\delta)}$$

such that

$$x = g_y(x).$$

But by Step 4, the equation

$$x = g_y(x)$$

is equivalent to

$$f(x) = y.$$

Therefore, for every

$$y \in \overline{B(0,\delta/2)},$$

there exists a unique

$$x \in \overline{B(0,\delta)}$$

such that

$$f(x) = y.$$

This proves that f has a local inverse near 0.

Step 8: Define the local inverse

Because for each small y there is a unique x satisfying

$$f(x) = y,$$

we may define

$$f^{-1}(y) = x.$$

This inverse is only local. It is defined for y in a sufficiently small neighborhood of 0.

Step 9: Prove that the inverse is continuous

Take two small output points

$$y_1, y_2.$$

Let

$$x_1 = f^{-1}(y_1), \qquad x_2 = f^{-1}(y_2).$$

Then

$$f(x_1) = y_1, \qquad f(x_2) = y_2.$$

Since

$$f(x) = x - g(x),$$

we have

$$y_1 = x_1 - g(x_1),$$

and

$$y_2 = x_2 - g(x_2).$$

Subtract the second equation from the first:

$$y_1 - y_2 = \big(x_1 - g(x_1)\big) - \big(x_2 - g(x_2)\big).$$

Distribute the minus sign:

$$y_1 - y_2 = (x_1 - x_2) - \big(g(x_1) - g(x_2)\big).$$

Now rearrange to isolate $x_1 - x_2$:

$$x_1 - x_2 = (y_1 - y_2) + \big(g(x_1) - g(x_2)\big).$$

Take norms:

$$\|x_1 - x_2\| \le \|y_1 - y_2\| + \|g(x_1) - g(x_2)\|.$$

Using the contraction estimate for g,

$$\|g(x_1) - g(x_2)\| \le \frac{1}{2}\|x_1 - x_2\|.$$

Therefore

$$\|x_1 - x_2\| \le \|y_1 - y_2\| + \frac{1}{2}\|x_1 - x_2\|.$$

Subtract $\frac{1}{2}\|x_1 - x_2\|$ from both sides:

$$\frac{1}{2}\|x_1 - x_2\| \le \|y_1 - y_2\|.$$

Multiply by 2:

$$\|x_1 - x_2\| \leq 2\|y_1 - y_2\|.$$

Since

$$x_i = f^{-1}(y_i),$$

this says

$$\|f^{-1}(y_1) - f^{-1}(y_2)\| \leq 2\|y_1 - y_2\|.$$

So f^{-1} is Lipschitz continuous, hence continuous.

Step 10: Prove that the inverse is differentiable

Let y be a small output point and let

$$x = f^{-1}(y).$$

Then

$$f(x) = y.$$

Now take a small increment h in the output space. Suppose

$$y + h$$

is still in the local output neighborhood. Define

$$x + k = f^{-1}(y + h).$$

This means

$$f(x + k) = y + h.$$

Since also

$$f(x) = y,$$

subtracting gives

$$f(x + k) - f(x) = h.$$

Because f is differentiable at x, we have the linear approximation

$$f(x + k) = f(x) + Df(x)k + r(k),$$

where the remainder satisfies

$$\frac{\|r(k)\|}{\|k\|} \to 0 \qquad \text{as } k \to 0.$$

Subtract $f(x)$ from both sides:

$$f(x + k) - f(x) = Df(x)k + r(k).$$

But we already know

$$f(x + k) - f(x) = h.$$

Therefore

$$h = Df(x)k + r(k).$$

For x sufficiently close to 0, the matrix $Df(x)$ is invertible. This is because $Df(0) = I$ is invertible, Df is continuous, and invertible matrices form an open set. Thus we may multiply by $[Df(x)]^{-1}$:

$$[Df(x)]^{-1}h = [Df(x)]^{-1}Df(x)k + [Df(x)]^{-1}r(k).$$

Since

$$[Df(x)]^{-1}Df(x) = I,$$

we obtain

$$[Df(x)]^{-1}h = k + [Df(x)]^{-1}r(k).$$

Rearrange:

$$k = [Df(x)]^{-1}h - [Df(x)]^{-1}r(k).$$

Now recall what k means:

$$k = (x + k) - x = f^{-1}(y + h) - f^{-1}(y).$$

Thus

$$f^{-1}(y + h) - f^{-1}(y) = [Df(x)]^{-1}h - [Df(x)]^{-1}r(k).$$

The first term

$$[Df(x)]^{-1}h$$

is linear in h. We must show that the second term is small compared to $\|h\|$.

Since $[Df(x)]^{-1}$ is a fixed matrix for this fixed point x, there is a constant $C > 0$ such that

$$\|[Df(x)]^{-1}v\| \leq C\|v\|$$

for every vector v. Therefore

$$\|[Df(x)]^{-1}r(k)\| \leq C\|r(k)\|.$$

We know

$$\frac{\|r(k)\|}{\|k\|} \to 0.$$

Also, from the Lipschitz estimate for the inverse,

$$\|k\| = \|f^{-1}(y + h) - f^{-1}(y)\| \leq 2\|h\|.$$

So as $h \to 0$, we also have $k \to 0$, and the error term becomes negligible compared to $\|h\|$. Hence

$$\frac{\|[Df(x)]^{-1}r(k)\|}{\|h\|} \to 0.$$

Therefore

$$f^{-1}(y + h) - f^{-1}(y) = [Df(x)]^{-1}h + o(\|h\|).$$

By the definition of derivative,

$$D(f^{-1})(y) = [Df(x)]^{-1}.$$

Since

$$x = f^{-1}(y),$$

we get

$$D(f^{-1})(y) = \left[Df(f^{-1}(y))\right]^{-1}.$$

This proves the special case.

Part II: The General Case

Now we prove the full theorem.

Let

$$f : U \subset \mathbb{R}^n \to \mathbb{R}^n$$

be C^1, let

$$p \in U,$$

and assume

$$Df(p)$$

is invertible.

The problem is that we may not have

$$p = 0, \qquad f(p) = 0, \qquad Df(p) = I.$$

So we convert the general case into the special case.

Step 1: Shift the input

Write points near p as

$$x = p + h.$$

Here h measures displacement from p:

$$h = x - p.$$

If $x = p$, then

$$h = 0.$$

So this change of variables moves the base point p to the origin.

Step 2: Shift the output

Consider

$$H(h) = f(p + h) - f(p).$$

Then

$$H(0) = f(p + 0) - f(p) = f(p) - f(p) = 0.$$

So this new function sends 0 to 0.

Now compute its derivative. Since

$$H(h) = f(p + h) - f(p),$$

and $f(p)$ is constant, we get

$$DH(h) = Df(p + h).$$

Therefore

$$DH(0) = Df(p).$$

Let

$$A = Df(p).$$

By hypothesis, A is invertible.

So

$$DH(0) = A.$$

But the special case requires the derivative at 0 to be I, not A.

Step 3: Normalize the derivative

Since

$$DH(0) = A,$$

near 0 the function H behaves like

$$H(h) \approx Ah.$$

To make the linear part look like h, we multiply by A^{-1}, because

$$A^{-1}(Ah) = h.$$

Therefore define

$$F(h) = A^{-1}H(h).$$

Since

$$H(h) = f(p+h) - f(p),$$

this means

$$F(h) = A^{-1}(f(p+h) - f(p)).$$

Now check the special-case assumptions.

First,

$$F(0) = A^{-1}(f(p) - f(p)) = A^{-1}0 = 0.$$

Second,

$$DF(h) = A^{-1}DH(h).$$

So

$$DF(0) = A^{-1}DH(0) = A^{-1}A = I.$$

Thus F satisfies

$$F(0) = 0, \qquad DF(0) = I.$$

So F is exactly in the special case already proved.

Step 4: Apply the special case to F

By the special case, F has a local inverse near 0. That means:

There exist neighborhoods of 0 such that for every small z, there is a unique small h satisfying

$$F(h) = z.$$

Equivalently,

$$h = F^{-1}(z).$$

Step 5: Translate the inverse of F back to the inverse of f

We now unwrap the definition of F.

Suppose

$$F(h) = z.$$

Using the definition of F,

$$A^{-1}(f(p+h) - f(p)) = z.$$

Multiply both sides by A:

$$f(p+h) - f(p) = Az.$$

Add $f(p)$ to both sides:

$$f(p+h) = f(p) + Az.$$

Now set

$$x = p + h.$$

Then

$$f(x) = f(p) + Az.$$

Now let

$$y = f(p) + Az.$$

We solve this equation for z. Subtract $f(p)$:

$$y - f(p) = Az.$$

Multiply by A^{-1}:

$$z = A^{-1}(y - f(p)).$$

Since

$$h = F^{-1}(z),$$

and

$$x = p + h,$$

we get

$$x = p + F^{-1}(z).$$

Substitute

$$z = A^{-1}(y - f(p)) :$$

$$x = p + F^{-1}\big(A^{-1}(y - f(p))\big).$$

But x is the unique point satisfying

$$f(x) = y.$$

Therefore

$$f^{-1}(y) = p + F^{-1}\big(A^{-1}(y - f(p))\big).$$

This formula shows explicitly how the local inverse of f is obtained from the local inverse of F.

Step 6: Derivative formula in the general case

We already know from the special case that local inverses satisfy the formula

$$D(F^{-1})(z) = \big[DF(F^{-1}(z))\big]^{-1}.$$

The construction of F involved only translations and multiplication by the invertible matrix A^{-1}, so the local invertibility and differentiability transfer back to f.

More directly, once f^{-1} exists and is differentiable, take

$$y = f(x).$$

Then

$$f^{-1}(f(x)) = x.$$

Differentiate both sides with respect to x. By the chain rule,

$$D(f^{-1})(f(x))Df(x) = I.$$

Multiply on the right by $[Df(x)]^{-1}$:

$$D(f^{-1})(f(x)) = [Df(x)]^{-1}.$$

Now write

$$y = f(x),$$

so

$$x = f^{-1}(y).$$

Therefore

$$D(f^{-1})(y) = \left[Df(f^{-1}(y))\right]^{-1}.$$

This proves the general inverse function theorem.

$$\boxed{D(f^{-1})(y) = \left[Df(f^{-1}(y))\right]^{-1}}$$

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